PROJECT PROPOSAL

Smart Academic Registration Advisor

A Rule-Based Decision-Support System for University Course Registration

Course: Software Engineering

Item	Details
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Faculty / Level	Computer Science (CS) – Third Year
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Submission date	.....
Deliverables	Task 1: Proposal (idea, problem, solution, team roles). Task 2: Requirements (user stories, non-functional) and Agile plan.

Contents

Part 1: Proposal

1. Project Overview
2. Problem Statement
3. Proposed Solution
4. Objectives
5. Scope and Limitations
6. Stakeholders and User Roles
7. Proposed Technologies
8. Team and Roles

Part 2: Requirements and Agile Plan

9. Functional Requirements (User Stories and Acceptance Criteria)
10. Non-Functional Requirements
11. Agile Methodology

Part 1: Proposal

1. Project Overview

The Smart Academic Registration Advisor is a web-based decision-support system that helps university students choose appropriate courses to register for each semester. It analyzes the student's academic record (cumulative GPA, passed courses and failed courses) and applies the faculty's official academic regulations.

The system does more than list available courses. It analyzes the student's situation, proposes a registration plan, and explains the reason behind every recommendation or restriction.

2. Problem Statement

Academic registration rules are complicated, and students commonly struggle with the following:

- Deciding which courses to register for in the current semester.
- Knowing whether failed courses must be retaken or can be postponed.
- Understanding how a low GPA affects registration eligibility and credit-hour limits.
- Identifying courses that have prerequisites.
- Understanding how postponing a course affects future semesters.
- Choosing between several possible registration plans without delaying graduation.

As a result, students make uninformed registration decisions, academic advisors are overloaded with routine questions, and graduation may be delayed unnecessarily.

3. Proposed Solution

A rule-based system built around a transparent Academic Decision Engine. The system works as follows:

1. Collects the student's academic information.
2. Analyzes the academic record and identifies completed and outstanding requirements.
3. Validates registration eligibility for each course (prerequisites, GPA restrictions, credit-hour limits, course availability and failed-course policy).
4. Classifies courses as Eligible, Recommended or Restricted, and explains each classification.
5. Generates one or more feasible registration plans and compares them.
6. Presents the plan to the student with explanations and refers exceptional cases to an academic advisor.

Important notes:

- All rules (GPA thresholds, credit-hour limits, failed-course policy) come only from the faculty's official regulations. No unverified rule is assumed.
- The system provides recommendations only. It does not enroll students automatically and does not guarantee a GPA improvement or a graduation date.

4. Objectives

- Reduce registration errors caused by misunderstanding academic regulations.
- Explain registration decisions to students clearly.

- Reduce the workload of academic advisors on routine cases.
- Deliver a system that is testable, maintainable and verifiable against the official regulations.

5. Scope and Limitations

In scope:

Student academic profile management, course catalog and prerequisite management, eligibility checking, academic record analysis, registration plan generation and comparison, academic rule management, and advisor review of exceptional cases.

Out of scope:

Attendance management, financial payments, library services, complaint management, general student services, and the actual enrollment of students in courses.

6. Stakeholders and User Roles

Role	Responsibilities
Student	Views academic information, eligible courses and recommendations; compares plans; understands restrictions.
Academic Advisor	Reviews student situations and proposed plans; handles exceptional cases.
Administrator	Manages courses, prerequisites, semester availability, academic rules and academic data.

7. Proposed Technologies

A web application, a relational database and a rule-based recommendation engine. Final technology choices will be made during the design phase.

8. Team and Roles

The project is carried out by one person, the student named on the cover page, who is responsible for all stages of the work. The engineering roles below are performed in turn according to the project phase:

Phase	Role	Responsibilities
Planning and tracking	Project Manager	Organizes tasks and schedule, tracks progress, compiles deliverables.
Requirements	Requirements Analyst	Extracts rules from the regulations and study plan; writes requirements and the SRS.
Design	System Designer	Creates UML diagrams and keeps them consistent with the requirements.
Development	Database and Backend Developer	Designs the ERD and database; implements the Academic Decision Engine.
Interfaces	Frontend Developer	Designs and builds the student, advisor and administrator interfaces.
Testing	Tester	Writes and executes test cases against the verified rules.

Part 2: Requirements and Agile Plan

9. Functional Requirements

Functional requirements are written as Agile user stories, each followed by its acceptance criteria.

9.1 Student

ID / Priority	User Story	Acceptance Criteria
FR-01 High	**As a** student, **I want to** view my academic profile (level, cumulative GPA, passed and failed courses), **so that** I can verify that my record is correct before registering.	• The profile shows level, cumulative GPA, passed courses and failed courses. • Missing or inconsistent record data is flagged with a clear message.
FR-02 High	**As a** student, **I want to** see which courses I am eligible to register for this semester, **so that** I know my available options.	• Each course is classified as Eligible, Recommended or Restricted. • Eligibility is based on prerequisites, course availability, GPA restrictions and the credit-hour limit. • The list is recalculated when the student record changes.
FR-03 High	**As a** student, **I want to** see the reason a course is restricted, **so that** I understand which rule applies and what I must do to become eligible.	• Each restricted course displays the applicable rule and its source in the faculty bylaws. • The message names the unmet prerequisite, GPA condition or credit-hour limit.
FR-04 High	**As a** student, **I want to** see how my failed courses are handled under faculty regulations, **so that** I know whether I must retake them and when.	• Failed courses are listed with the applicable policy defined in the faculty bylaws. • The system does not assume that every failed course must be retaken first; each course is evaluated by its role in the curriculum.
FR-05 High	**As a** student, **I want to** receive ranked course recommendations with explanations, **so that** I can focus on the courses that matter most for my progress.	• Courses are ranked using documented criteria (mandatory courses, failed courses, prerequisites for future courses, risk of delay). • Each recommendation shows course name and code, credit hours, priority and reason. • The consequence of postponing a course is shown when it can be established from the study plan.
FR-06 High	**As a** student, **I want to** receive a proposed registration plan that respects my credit-hour limit, **so that** I can register with confidence.	• Total credit hours do not exceed the limit defined in the faculty bylaws. • Every course in the plan satisfies its prerequisites and is available this semester.

ID / Priority	User Story	Acceptance Criteria
FR-07 Medium	**As a** student, **I want to** compare alternative registration plans, **so that** I can choose the plan that best fits my situation.	• At least two plans are shown when more than one valid plan exists. • The comparison lists the advantages and disadvantages of each plan. • The system does not guarantee a GPA improvement or a graduation date.
FR-08 Medium	**As a** student, **I want to** be told when my case requires academic advisor review, **so that** I seek the right approval before registering.	• Exceptional or ambiguous cases are flagged automatically. • The plan status is shown as "Pending advisor review".

9.2 Academic Advisor

ID / Priority	User Story	Acceptance Criteria
FR-09 Medium	**As an** academic advisor, **I want to** review the students assigned to me together with their proposed plans, **so that** I can verify that the recommendations are appropriate.	• The advisor sees only the students assigned to them. • The advisor can view the student record, the proposed plan and the reasons behind it.
FR-10 Medium	**As an** academic advisor, **I want to** approve or comment on plans for exceptional cases, **so that** students receive an authorized decision.	• The advisor can approve or add a comment when faculty policy permits. • The decision is saved with the advisor name and date and is visible to the student.

9.3 Administrator

ID / Priority	User Story	Acceptance Criteria
FR-11 High	**As an** administrator, **I want to** manage the course catalog (code, name, credit hours, mandatory or elective, level), **so that** the system always reflects the official study plan.	• Courses can be added, edited and deactivated. • A duplicate course code is rejected with an error message.
FR-12 High	**As an** administrator, **I want to** manage the prerequisites of each course, **so that** eligibility checks are accurate.	• A prerequisite can be linked to or removed from a course. • A circular prerequisite chain is rejected.
FR-13 High	**As an** administrator, **I want to** set which courses are offered in each semester, **so that** plans include only courses that are actually available.	• Courses not offered in the selected semester never appear in eligible lists or plans.

ID / Priority	User Story	Acceptance Criteria
FR-14 High	**As an** administrator, **I want to** maintain the verified academic rules (GPA thresholds, credit-hour limits, failed-course policy), **so that** the decision engine always applies the current faculty regulations.	• Each rule stores its value and its source (bylaw section or page). • A changed rule is applied to all new calculations.
FR-15 Medium	**As an** administrator, **I want to** view a history of changes made to academic rules, **so that** changes are traceable and accountable.	• Each change records the user, date and time, old value and new value.
FR-16 High	**As an** administrator, **I want to** enter or import student academic records, **so that** the system has the data it needs to analyze each student.	• Required fields are validated before saving. • Invalid or incomplete records are rejected with a clear error message.

9.4 All Users

ID / Priority	User Story	Acceptance Criteria
FR-17 High	**As an** user (student, advisor or administrator), **I want to** log in with my role-specific account, **so that** I can access only the features and data I am permitted to use.	• The system supports three roles: Student, Academic Advisor and Administrator. • A student can view only their own data. • Unauthorized access attempts are denied.

10. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	"All student academic data must be protected: passwords must be stored encrypted, and access must be controlled by user role so that a student can view only their own data and an advisor only the students assigned to them."
NFR-02	Accuracy and Transparency	"Eligibility decisions must match the verified faculty regulations in 100% of the approved test cases, and every recommendation or restriction must display a written reason that refers to the rule applied."
NFR-03	Usability	"The interface must be mobile-responsive and understandable by a student without prior training."
NFR-04	Maintainability	"Academic rules must be stored as editable data, separate from the application code, so that rule changes do not require rewriting the decision engine."

11. Agile Methodology

11.1 Why Agile?

Academic rules are confirmed gradually as the faculty regulations are studied, so requirements may change. Agile allows the system to be delivered and reviewed in small increments instead of waiting until the end of the project.

11.2 Adaptation for a One-Person Project

- The project owner acts as both Product Owner and Developer instead of using the full Scrum roles.
- Short Sprints (two weeks, adjusted to the submission deadline).
- A brief self-review (Review and Retrospective) at the end of each Sprint.
- A Product Backlog of user stories ordered by priority, kept on a task board (for example Trello or GitHub Projects).

11.3 Product Backlog Priorities

Priority	User Stories
High	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-11, FR-12, FR-13, FR-14, FR-16, FR-17
Medium	FR-07, FR-08, FR-09, FR-10, FR-15

11.4 Proposed Sprint Plan

Sprint	Goal	Deliverables and Stories
0	Preparation	Obtain the regulations and study plan, extract the rules, set up the Product Backlog.
1	Requirements and design	SRS, Use Case Diagram, ERD and database design.
2	Foundation	Database, login and roles (FR-17), course catalog (FR-11), prerequisites (FR-12), semester availability (FR-13), student records (FR-16).
3	Decision engine (1)	Rule management (FR-14), eligibility checking (FR-02), restriction reasons (FR-03), failed-course handling (FR-04).
4	Decision engine (2)	Prioritization and recommendations (FR-05), plan generation (FR-06), plan comparison (FR-07).
5	Interfaces and review	Student profile (FR-01), advisor flow (FR-08, FR-09, FR-10), rule change history (FR-15).
6	Testing and delivery	Execute test cases, fix defects, final documentation.

11.5 Definition of Done

A user story is considered done when:

7. It is implemented and works correctly.
8. All of its acceptance criteria are met and tested.
9. It matches a documented rule from the faculty regulations.
10. Related documents (requirements, diagrams) are updated.